## Path-Guided Reasoning (PGR) vs. Flat Retrieval — Bridge Query

Query: *"How does Varroa Destructor connect to Polyface?"* Bridge entity: **Honey Bees**

### GRASP-RAG (bridge motif + PGR context)

*Context sent to LLM (built by `_bridge_context()`):*

Parasitic mite that infests honey bee colonies; causes parasitic mite syndrome and high colony mortality. . .
*[source passage]*

**Keywords:** colony infestation, bee mortality, mite resistance
*[source passage]*

Central to pollination of crops; managed at Polyface Farm as part of integrated sustainable operations. . .
*[source passage]*

**Keywords:** pollination services, sustainable agriculture, local food systems
*[source passage]*

Joel Salatin's regenerative farm; integrates multiple species and relies on pollination for crop production. . .

*Answer generated from context above (real output):*

**Answer:** "Varroa Destructor. . . infests **honey bee colonies**. . . healthy bee populations are essential. . . for the pollination of various crops, which Polyface incorporates into its local food systems." ✓ **bridge entity named**

### LightRAG / PathRAG (flat retrieval)

Passage 1: Varroa Destructor is a major mite pest of honey bee colonies; causes parasitic mite syndrome. . .
Passage 2: Polyface Farm is Joel Salatin's sustainable farm in Virginia; known for multi-species rotational methods. . .

*keywords: sustainable agriculture, pest management, beekeeping, regenerative farming. . .*

*No path between anchors. Varroa Destructor and Polyface retrieved independently — the connecting node (Honey Bees) is not labeled or surfaced.*

*Real outputs from both systems:*

**LightRAG:** "I do not have enough information to answer that."
**PathRAG:** "I'm sorry, but I cannot provide information connecting Varroa Destructor to Polyface. . ."
× **bridge entity not found**

**Why:** Flat retrieval finds passages *about* each anchor separately; without a graph path, the connecting node (Honey Bees) is never retrieved. GRASP-RAG's bidirectional shortest-path guarantees traversal of the bridge node when both anchors are in the graph. Win rate on bridge queries vs. LightRAG: **93.4%**.